Final Exam Cheat Sheet — AWS AIP-C01

Exam: **Mon 9 Nov 2026, 1:30 PM IST** · Built by Theo prof 2 from Days 1–9, the gap lesson, End Game Day, both practice sets and every quiz · All 98 official skills taught

1. Exam-day strategy

- Format: 75 questions (65 scored + unscored ones that look identical), multiple choice (1 of 4) and multiple response (2+ of 5–7, all-or-nothing). Scaled score 100–1000; professional exams usually pass at 750.
- **Answer every question** — blank = wrong, no penalty for guessing. Flag and move on after ~3 minutes.
- Pace: ~2.5 min per question → question 25 by ~2:35 PM, question 50 by ~3:40 PM, leave 15+ min to review flags.
- **List every requirement in the stem** (cost, latency, 'no code changes', 'least overhead', residency, audit…) and strike any option that fails even one.
- **Read every option to the last word** — look-alike options differ in one phrase (Express vs Standard, draft vs alias, CloudTrail vs invocation logs).
- Multi-finding stems: map each finding to an option before submitting; a missing fix makes the answer wrong.
- Wrong options are usually real AWS approaches that fail one requirement — ask 'which requirement does this miss?'

2. Decision rules by domain — 'stem says → answer'

Domain 1 — FM integration, data, compliance (31%)

Stem says…	Answer
Knowledge changes often / must cite sources	RAG (Knowledge Base) + scheduled or event-driven sync
Style, tone, format, domain behaviour	Fine-tuning (labelled pairs); raw domain text → continued pre-training
Small model must behave like a big one	Distillation; trained elsewhere → Custom Model Import
Most cost-effective model meeting quality	Smallest adequate model, proven with Bedrock model evaluation
Consistent output	Low temperature; correctness → grounding; cut off → raise maxTokens
Custom logic between retrieval and generation	Retrieve + Converse (not RetrieveAndGenerate)
Exact codes/IDs missed	Hybrid search; right docs wrong order → reranker; context lost → hierarchical chunking
Tables / complex PDFs parsed badly	FM parsing or Bedrock Data Automation
Relationships across entities	GraphRAG (Neptune Analytics)
Vector store options	OpenSearch Serverless, Aurora pgvector, Neptune Analytics, S3 Vectors, Pinecone, MongoDB Atlas, Redis — never DynamoDB
Vague / multi-part / follow-up questions	Query expansion / decomposition / rewriting to standalone
Rule-based data checks at scale, no code	Glue Data Quality (DQDL) → EventBridge halt
Visual dataset prep for training	SageMaker Data Wrangler
Many media types → one schema, managed	Bedrock Data Automation; calls → Transcribe; scans → Textract; custom container → SageMaker Processing
Switch models with one request format	Converse API
Persona/rules consistent across teams, change without redeploy	Prompt Management versions referenced by ARN (pointer in AppConfig)
Parser-safe structured output	Tool schema + toolChoice + Lambda validation
Prompt chain with branching inside Bedrock, low-code	Bedrock Flows (alias + InvokeFlow)
Missing details must be asked, long waits	Step Functions Standard + .waitForTaskToken; history in DynamoDB
New prompt must not reduce quality	Golden set + Map state + Lambda checks + PassRate alarm gate
Validate feasibility before investing	PoC: measurable criteria first, on-demand, sample KB, evaluation jobs
Consistent implementation across teams	WA Tool + GenAI Lens → reusable CDK/CloudFormation via Service Catalog

Domain 2 — Implementation and integration (26%)

Stem says…	Answer
Fixed, known steps	Workflow / Step Functions / Flows — not an agent
Path depends on intermediate results	Agent (ReAct loop with iteration cap)
Full control of the tool loop	Converse API with toolConfig; results as toolResult in a user message
Framework-agnostic agent hosting, session isolation	AgentCore Runtime
Reuse tools across teams and frameworks	MCP, often via AgentCore Gateway
Act in third-party apps as the user	AgentCore Identity (outbound OAuth)
Remember users across sessions	AgentCore Memory long-term
Each request to one specialist / several combined	Router (Agent Squad classifier) / supervisor
Runaway agents	Stopping conditions, iteration caps, timeouts, budgets, least-privilege IAM
Stop hammering a failing model	Circuit breaker (Step Functions + DynamoDB state)
Switch model without redeploy	AppConfig (+ gradual rollout, alarm rollback)
Route by content to specialised FMs	Classify step + Choice state; same family by quality → Intelligent Prompt Routing
Throttling	Backoff + jitter → cross-Region inference → SQS buffer → quota / Provisioned Throughput
User waiting, feels slow	Streaming (ConverseStream → WebSocket / SSE)
API Gateway 504 at ~29 s	Stream, or async: job ID + SQS + worker + stored result
Huge non-urgent volume	Batch inference (~50% cheaper)
Legacy SOAP / fragile on-prem system	API Gateway throttling + Lambda adapter + VPN/Direct Connect (+ SQS buffer)
Webhook with short sender timeout	Verify, enqueue, 200 fast; worker does the GenAI work
Only documents the user may see	Identity group claim → KB metadata filter
Data must stay on premises / 5G edge latency	Outposts / Wavelength
Many teams: approved models, quotas, cost per team, swap providers	GenAI gateway + usage plans + application inference profiles
Ship GenAI changes safely	CodePipeline + CodeBuild (prompt regression, guardrail, security tests) + CodeDeploy canary with alarm rollback
Chat UI fast / API contract first / speed up developers	Amplify UI / OpenAPI / Amazon Q Developer

Domain 3 — Safety, security, governance (20%)

Stem says…	Answer
Must always block, regardless of user input	Guardrails (not the system prompt)
A subject however worded / exact words	Denied topic / word filter
Answer must be supported by sources	Contextual grounding (grounding + relevance thresholds)
Provably consistent with written rules	Automated Reasoning checks
Same policies for non-Bedrock models	ApplyGuardrail; only user text → input tagging
Hidden instructions in documents/emails	Indirect prompt injection: sanitise, check retrieved content, least privilege, human confirmation
Find PII in S3 / redact in pipeline / mask at runtime	Macie / Comprehend / Guardrails
Column or row restrictions on lake data	Lake Formation
No public internet	Interface VPC endpoints (bedrock-runtime, bedrock-agent-runtime)
Restrict models for a team / whole org	IAM with model ARNs / SCP deny
Own and rotate keys; sync fails after enabling KMS	Customer managed KMS key; key policy must allow the role
Who did it / what text / why this answer	CloudTrail / invocation logging / decision logs
Must not change for N years	S3 Object Lock (compliance mode)
Fairness	Per-group metrics + counterfactual tests; document limits in model cards

Domain 4 — Efficiency and monitoring (12%)

Stem says…	Answer
Same long prefix every call	Prompt caching (static content first)
Identical / reworded repeat questions	Exact-match cache / semantic cache (invalidate on KB sync)
Latency dominated by long answers	Cap output length first
Cost per team	Application inference profiles + tags
Steady guaranteed capacity / spiky	Provisioned Throughput / on-demand (+ cross-Region inference)
Metric varies by time of day	CloudWatch anomaly detection
Unexpected spend	AWS Cost Anomaly Detection
Which service in the chain is slow	X-Ray
Analyse prompts/responses at scale	Logs Insights / Athena over invocation logs
Agent cost up, answers fine	Tool call baselines → loops
Compliance drift (logging off, no guardrail)	AWS Config rules
Vector store slow / stale / deleted docs still found	Index tuning, capacity, filters / sync deletions

Domain 5 — Testing, validation, troubleshooting (11%)

Stem says…	Answer
Pick a model with our own data, least effort	Bedrock model evaluation job (automatic / LLM-as-a-judge / human)
Is the retriever the problem?	Retrieve alone; KB evaluation retrieve-only (context relevance, coverage)
Validate on real traffic, no user impact	Shadow testing
Gradual release with automatic rollback	Canary + alarms
Compare business outcomes of two configs	A/B test
Broken flow when no users are active	CloudWatch Synthetics canaries
Meaning of answers changed after upgrade	Semantic drift check vs baseline
Same question, different answers	Output diffing / consistency analysis (+ temperature)
Right data, wrong conclusion	Reasoning path tracing
Block deployments that reduce quality	Automated quality gate in the pipeline
Scale + subjective tone + learn from users	LLM-as-a-judge + human annotation sample + user feedback
Wrong behaviour, latency normal	Rendered prompt in invocation logs (Logs Insights)
stopReason max_tokens, small input	Raise maxTokens / shorter format
ValidationException / AccessDenied	Fix, don't retry; Throttling / ServiceUnavailable → retry

3. Look-alike pairs the exam uses to confuse you

Pair	The difference
Grounding vs filtering	Answer true to sources (Guardrails) vs who may see which sources (metadata filter)
CloudTrail vs invocation logs vs decision logs	Who called which API vs exact prompt/response text vs why this answer (sources, versions)
X-Ray vs Logs Insights	Where time/errors occur across services vs what the content was
Retrieve vs RetrieveAndGenerate	Chunks only (custom logic after) vs managed retrieval + answer
Step Functions Express vs Standard	Max 5 minutes, high volume vs up to 1 year, waits, full history
Batch inference vs async SQS vs StartAsyncInvoke	Managed file job, hours, ~50% cheaper vs user waiting, on-demand price vs native async for slow media generation
Provisioned Throughput vs cross-Region inference	Guaranteed capacity, commitment vs more throughput, no extra cost
Intelligent Prompt Routing vs content routing	Within one model family by predicted quality vs classify + Choice
Denied topics vs word filters	Meaning vs exact words
Contextual grounding vs Automated Reasoning	Similarity to sources vs formal logic against rules
Fine-tuning vs continued pre-training vs distillation	Labelled pairs vs raw text vs teacher → student
Prompt Management vs Flows/Agents lifecycle	Versions only (pointer in config) vs versions + aliases
Prompt caching vs exact cache vs semantic cache	Reused prefix tokens vs identical request vs similar meaning

Pair	The difference
Hybrid search vs reranker vs hierarchical chunking	Exact terms vs ordering vs lost context
Temperature vs top-p vs top-k	Randomness vs cumulative probability threshold vs fixed count
Router vs supervisor	One specialist per request vs combining several
Macie vs Comprehend vs Guardrails (PII)	Find in S3 vs redact in pipeline vs mask/block at runtime
Glue Data Quality vs Data Wrangler	Automated rules in pipelines vs visual exploration
BDA vs Textract vs SageMaker Processing	Many media → schema vs document OCR vs custom container
Outposts vs Wavelength vs Local Zones	Your data centre vs 5G edge vs metro site
Q Developer vs Q Business	Developers' coding assistant vs employees' assistant over company data
A/B vs canary vs shadow	Compare outcomes vs safe gradual rollout vs no user impact
Anomaly detection vs Cost Anomaly Detection vs Budgets	Metric band vs unusual spend vs fixed amount
Replicas vs shards	Throughput + availability vs parallel capacity for big indexes
Agent action group vs MCP / AgentCore Gateway	Tools for one Bedrock agent vs tools shared across frameworks

4. Trick-question cues

Cue in the stem or option	What it signals
'LEAST operational overhead'	Managed feature (KB, Flows, Guardrails, BDA) beats custom Lambda/EC2 code
'MOST cost-effective'	Smallest adequate model, batch, caching, routing — not Provisioned Throughput by reflex
'without code changes / redeploying'	AppConfig, Prompt Management version pointer, aliases, gateway
'always / regardless of input / enforce'	Guardrails, IAM, SCP — never a prompt instruction
'only the documents they are allowed to see'	Metadata filter at retrieval from identity claims
'exact text' vs 'who'	Invocation logging vs CloudTrail
'reworded' / 'similar questions'	Semantic, not exact-match, cache
'hours' / 'days' / 'human approval'	Step Functions Standard + task token, not Express
'no impact on users'	Shadow testing
'real-time' / 'user waiting'	Streaming or synchronous; 'overnight' → batch
Option fine-tunes for facts that change	Wrong — RAG
Option retries a ValidationException	Wrong — fix the request
Option puts security in the system prompt	Wrong — Guardrails / IAM / filters
Option uses DynamoDB as the vector store	Wrong — metadata or history only
'publish a wiki' / 'ask teams to' / 'review manually monthly'	Wrong when consistency or automation is required
Option correct but covers only part of the stem	Wrong — your most common past error pattern
Draft (agent, flow, prompt) used in production	Wrong — versions/aliases

5. Numbers and exact names

- Step Functions Express ≤ 5 min; Standard ≤ 1 year. Lambda ≤ 15 min. API Gateway integration timeout 29 s default.
- Batch inference ≈ 50% cheaper than on-demand; input is JSONL (recordId + modelInput).
- KB 'data source' (not data store); endpoint **bedrock-agent-runtime** for Retrieve, InvokeAgent, InvokeFlow; **bedrock-runtime** for Converse / InvokeModel.
- 'Amazon Bedrock Flows' (formerly Prompt Flows). Model lifecycle Active → Legacy → End-of-Life.
- Tool results: toolResult blocks in ONE role 'user' message with the matching toolUseId.
- stopReason values: end_turn, max_tokens, tool_use, guardrail_intervened, content_filtered.
- Bedrock CloudWatch metrics: Invocations, InvocationLatency, InvocationClientErrors, InvocationServerErrors, InvocationThrottles, InputTokenCount, OutputTokenCount.

6. Your mistakes log — read this last tonight and first tomorrow

- DynamoDB is NOT a KB vector store (OpenSearch Serverless, Aurora pgvector, Neptune Analytics, S3 Vectors, Pinecone, MongoDB Atlas, Redis).
- Tool results → toolResult blocks in ONE role "user" message (matching toolUseId); never in the system prompt.
- Custom logic between retrieval and generation → Retrieve + Converse (not RetrieveAndGenerate, not an agent).
- Step Functions Express max 5 min; Standard up to 1 year; approval timeout → TimeoutSeconds/HeartbeatSeconds + Catch States.Timeout.
- Grounding (answer faithful to sources → Guardrails contextual grounding) vs filtering (who sees which docs → metadata filter at retrieval).
- Exact names: KB "data source" (not data store); "Amazon Bedrock Flows"; endpoint "bedrock-agent-runtime".
- KMS AccessDenied on KB sync → key policy must allow the KB role kms:Decrypt.
- Top-p = cumulative probability threshold (top-k = fixed count). Temperature → consistency; grounding → correctness.
- Batch inference (managed file job, hours, ~50% cheaper) vs async SQS (user waiting, on-demand price).
- Cascading (small first, escalate on failed validation) keeps quality on hard queries; distillation alone may not.
- Intelligent Prompt Routing = within one model family by predicted quality; business-category routing = classify step + Choice state.
- Different frameworks + consistent auth/format for retrieval → one MCP tool server, not direct per-team access.
- Fix (don't retry): ValidationException, AccessDeniedException. Retry: Throttling, ServiceUnavailable, ModelNotReady.
- Reworded repeat questions → semantic cache (exact-hash only hits identical text); invalidate on KB sync.
- Cost per success: cheap model + escalation = price_B + failure_rate × price_A; compare with price_A / success_A.
- Latency dominated by long generation → cap output length first.
- Per-answer audit (sources, model + guardrail versions, decision) → decision logs in CloudWatch Logs; model cards = model-level docs; Glue lineage = dataset origins.
- Wrong behaviour with normal latency → rendered prompt in invocation logs (Logs Insights); X-Ray = where time/errors occur.

- Multi-finding stems (Select ALL): map each finding to its fix (hard-coded prompts → Prompt Management templates, IAM-gated versions).
- Habit: list every requirement in the stem and strike options that fail any one.
- Habit: read every option to the last word; flag look-alike options.

7. Consolidated glossary — 353 terms (all days, alphabetical)

Term	Meaning
.waitForTaskToken	Step Functions callback: pause until a token is returned with a result.
A/B test / canary / shadow test	Compare configs on split users / small % rollout / mirror traffic without showing results.
A2A (Agent-to-Agent protocol)	Open standard for agents to communicate with other agents.
Action group	Bedrock Agents Classic's way of defining tools (OpenAPI schema or function details).
Active / Legacy / End-of-Life	Bedrock model lifecycle stages.
Adapter (Lambda)	Translator between a modern JSON API and a legacy protocol such as SOAP.
Adversarial testing / red-teaming	Running attack prompts systematically, e.g. on every release.
Agent	A model plus instructions, tools and memory running in a loop, deciding at runtime which tool to call.
Agent Service (Foundry Agent Service)	Azure's managed agent runtime (threads, messages, runs).
Agent Squad (AWS Agent Squad (formerly Multi-Agent Orchestrator))	Open-source framework that routes each request to the best specialist agent.
AgentCore (Amazon Bedrock AgentCore)	AWS services to host, connect, secure and observe agents built with any framework.
Agents Classic (Amazon Bedrock Agents (Classic))	The original managed agents service, now in maintenance mode for new accounts.
AI (Artificial Intelligence)	Software that performs tasks needing human-like judgement.
AI Search (Azure AI Search)	Azure's search/vector service.
AI-103	Microsoft exam: Developing AI Apps and Agents on Azure.
AI-500	Microsoft exam: Designing and Implementing Multi-Agent AI Solutions (Expert).
AIP-C01	Exam code for AWS Certified Generative AI Developer – Professional.
ALB (Application Load Balancer)	Distributes HTTP traffic across targets.
Amazon A2I (Augmented AI)	Managed human review workflows for ML outputs.
Amazon AppFlow	Managed no-code data transfer from SaaS apps to AWS.
Amazon Bedrock Flows	Visual chaining of prompt, condition, KB, Lambda and other nodes; versions + aliases.
Amazon Bedrock Guardrails	Managed safety layer checking model input and output against your policies.
Amazon Bedrock Prompt Management	Managed, versioned store for prompt templates, model and settings.
Amazon Cognito user / identity pool	App user sign-in and tokens / exchange for temporary AWS credentials.
Amazon EventBridge	Event bus routing events to targets by rules.
Amazon Q Business	Enterprise assistant answering employees from company data.
Amazon Q Developer	AI coding assistant: generate, refactor, test, explain, optimise code.

Term	Meaning
Amazon Textract	OCR service extracting text, key-value pairs and tables from documents.
Amazon Transcribe	Speech-to-text with diarization, custom vocabulary and PII redaction.
ANN / HNSW / ef_search	Approximate nearest-neighbour search; HNSW algorithm; recall-vs-latency setting.
API (Application Programming Interface)	A defined way for programs to call each other.
API Gateway (Amazon API Gateway)	Managed front door for HTTP/REST APIs.
API Gateway request validation	Schema checks that reject malformed requests before the backend runs.
API Gateway timeout	Default 29-second integration timeout for synchronous APIs.
Application inference profile	Your own tagged profile to track cost/usage per team.
ARN (Amazon Resource Name)	The unique ID string of an AWS resource.
Aurora / pgvector (Amazon Aurora PostgreSQL + pgvector extension)	Relational database with vector search support.
Automated Reasoning checks	Formal-logic verification of answers against rules from a policy document.
Automated remediation	Alarm → EventBridge → Lambda/SSM action without human delay.
AWS (Amazon Web Services)	Amazon's cloud platform.
AWS Amplify / Amplify UI	Front-end framework and hosting / declarative ready-made UI components.
AWS AppConfig / feature flag	Service for runtime settings; a flag is an on/off or value read while the app runs.
AWS Config	Continuous checks of resource configuration against rules.
AWS Config / AWS Audit Manager	Configuration compliance rules / continuous audit evidence collection.
AWS Cost Anomaly Detection	Billing feature alerting on unusual spend patterns.
AWS DMS / CDC	Database Migration Service / Change Data Capture streaming only changed rows.
AWS Glue Data Catalog	Central inventory of datasets and schemas.
AWS Glue Data Quality / DQDL	Rule-based data checks in Glue / the rule language (Completeness, Uniqueness…).
AWS Local Zones	Small AWS sites near large cities.
AWS Organizations	Groups many AWS accounts under central management; where SCPs live.
AWS Outposts	AWS-managed racks running AWS services in your own data centre.
AWS Service Catalog	Catalogue of pre-approved products teams can launch themselves.
AWS Wavelength	AWS compute inside telecom 5G networks for ultra-low latency.
AWS Well-Architected Framework	Best-practice questions in six pillars.
AWS X-Ray	Distributed tracing: time and errors per service segment of one request.
AWS X-Ray / service map	Distributed tracing with per-step timings; diagram of services.
aws:SourceVpce	IAM condition key: which VPC endpoint the request came through.
Batch inference / JSONL	File-based bulk job (~50% cheaper); JSON Lines = one JSON object per line.
BDA (Amazon Bedrock Data Automation)	Managed extraction of structured data from documents, images, audio, video.
Bedrock CloudWatch metrics	Invocations, InputTokenCount, OutputTokenCount, InvocationLatency, InvocationThrottles.

Term	Meaning
Bedrock Data Automation (BDA) / blueprint	Managed extraction from documents and media / your field schema for it.
Bedrock Marketplace	Additional specialised models on managed endpoints via Bedrock APIs.
Bedrock model evaluation	Managed comparison of models on your dataset (automatic, LLM-as-a-judge, human).
Bedrock model evaluation job	Managed comparison of models on a dataset (automatic, judge, human).
BLEU / ROUGE / BERTScore	Word-overlap metrics / meaning-based similarity metric.
Blue/green, canary, linear	Deployment strategies shifting traffic gradually; auto-rollback on alarms.
BM25 (Best Matching 25)	A classic keyword-ranking algorithm (exact word matching).
boto3	The AWS SDK for Python.
Capacity planning	Estimating needed tokens/requests per minute from peak traffic.
Cascading	Try a small model first; escalate to a larger one if quality checks fail.
CDK (AWS Cloud Development Kit)	Define infrastructure in code (TypeScript, Python…).
CDK / CloudFormation	Infrastructure as code in Python/Java / AWS template service.
Chain-of-thought (CoT)	Model reasons step by step before the final answer.
Chunked transfer encoding	HTTP response sent in pieces.
Chunking	Splitting documents into smaller passages before embedding.
CI/CD	Continuous Integration / Continuous Delivery: automated build, test, deploy.
Circuit breaker (closed/open/half-open)	Stops calls to a failing service, uses a fallback, tests recovery.
Clarification workflow	Workflow that asks the user for missing details before acting.
CLI (Command Line Interface)	Terminal tool for calling a service.
CloudShell (AWS CloudShell)	Browser terminal with the AWS CLI pre-configured.
CloudTrail (AWS CloudTrail)	Records who called which AWS API, and when.
CloudTrail Lake	Queryable (SQL) store of CloudTrail audit events.
CloudWatch (Amazon CloudWatch)	Metrics, logs, dashboards and alarms.
CloudWatch anomaly detection	Learned expected band for a metric; alarm when outside it.
CloudWatch Logs Insights	Query language over CloudWatch log data.
CloudWatch Synthetics canary	Scheduled script acting as a synthetic user against an endpoint.
CMK / customer managed key (Customer managed KMS key)	A KMS key you create and control (rotation, policy, revocation).
Code Interpreter	A sandbox where an agent runs code it writes.
CodePipeline / CodeBuild / CodeDeploy	Pipeline orchestration / build and test runs / gradual deploys with rollback.
Cognito (Amazon Cognito)	User sign-up/sign-in; issues JWTs.
Comprehend (Amazon Comprehend)	Natural-language service; detects and redacts PII.
Concurrency control / reserved concurrency	Capping simultaneous calls; Lambda's per-function concurrency setting.
Condition node / InvokeFlow	Flow branching on expressions / API that runs a flow (via an alias).
Confidence scoring	Estimating certainty, e.g. agreement across repeated answers.
Content filters / filter strength	Harm-category detection; None–High sensitivity per input/output.

Term	Meaning
Content Safety (Azure AI Content Safety)	Azure content filtering incl. groundedness detection.
Context precision / recall	Precision: are retrieved chunks relevant? Recall: was everything needed retrieved?
Context pruning / sliding window	Dropping or summarising old conversation turns to cut tokens.
Context relevance / context coverage	Retrieved chunks are relevant / contain the needed information.
Context window	The maximum number of tokens a model can consider in one request.
Contextual grounding check	Grounding and relevance scores of an answer against source and question.
Continued pre-training	Further training on unlabelled domain text to learn vocabulary.
Converse / ConverseStream	Unified Bedrock call format across models; streaming variant.
Correlation ID	Request ID carried through every log line to trace one request end to end.
Cost per successful task	Total cost including failures, retries and escalations divided by successes.
Counterfactual testing	Same request with only a sensitive attribute changed.
CRIS (Cross-Region Inference)	Bedrock routes requests to other regions for capacity.
Cross-region inference / inference profile	Routing requests across Regions in a geography via a profile ID (us., eu., apac., global.).
Custom Model Import	Bring open-weight model weights from S3 into Bedrock, served serverlessly.
Custom scoring / boosting	Adjusting search ranking by weights or fields such as recency.
Data lineage	Record of where data came from and how it was transformed.
Data residency	Requirement that data stays in a specific country/region.
Data source	The location a Knowledge Base ingests from (e.g. an S3 bucket); chunking is set here.
Decision log	Per-answer record of inputs, sources, versions, output and decision.
Defence in depth	Several independent protective layers.
Denied topics	Subjects defined in plain English, blocked by meaning.
Direct / indirect prompt injection	Malicious instructions typed by the user / hidden in content the model reads.
Direct Connect / Site-to-Site VPN	Dedicated private line / encrypted tunnel between on-premises and AWS.
DNS / private DNS (Domain Name System)	Translates service names into addresses; private DNS makes the normal service name resolve to your endpoint.
Draft / prompt version	Editable working copy / immutable numbered snapshot used in production.
DRAFT / version / alias	DRAFT is the editable agent; Prepare → version (snapshot); an alias points to a version.
Drift / bias drift	Gradual change in data or behaviour / in fairness metrics over time.
Dynamic chunking	Choosing chunking strategy and size per document type.
DynamoDB (Amazon DynamoDB)	Key-value / document NoSQL database. **Not** a KB vector store.
ECS / Fargate (Elastic Container Service / serverless containers)	Run containers; Fargate removes server management.
Embedding	A list of numbers representing the meaning of text; similar meanings give nearby vectors.
Endpoint policy	A policy on a VPC endpoint limiting what can be called through it.
Ensemble / aggregation	Several models answer; combine by vote, judge, merge.

Term	Meaning
Entra ID (Microsoft Entra ID)	Azure's identity service (formerly Azure AD).
ESL +30 (English as a Second Language +30 minutes)	AWS exam accommodation (not requested).
ETL (Extract, Transform, Load)	Moving and reshaping data between systems.
EventBridge (Amazon EventBridge)	Event bus and scheduler that triggers targets on events or schedules.
Exact-match cache / request hashing / fingerprint	Cache keyed by a hash of the normalised request.
Exponential backoff	Retry with growing waits (1s, 2s, 4s…).
Exponential backoff + jitter	Retries with growing, slightly randomised waits.
Extended thinking	Built-in reasoning mode with a token budget on some models.
Fairness metric	Measure compared across groups (refusal rate, quality, sentiment).
Faithfulness	Every claim in the answer is supported by the provided context.
Feedback loop	Collect user/validator signals → improve prompt → test → new version.
Few-shot prompting	Including a few example input/output pairs in the prompt.
Fine-tuning	Further training a model on your examples to change its behaviour or style.
FM (Foundation Model)	A large, general-purpose pre-trained model you build applications on.
Format drift	Output structure gradually deviating from the expected schema.
Foundation model (FM)	Large pre-trained general-purpose model you build on without training it yourself.
Foundry (Microsoft (Azure AI) Foundry)	Azure's platform for models and agents.
GenAI (Generative AI)	AI that creates new content (text, images, code) instead of only classifying.
GenAI gateway	Central layer all apps use to reach models: auth, quotas, routing, guardrails, logging, cost.
Generative AI Lens / WA Tool	GenAI-specific question set / console service running reviews with milestones.
Glue / Data Catalog (AWS Glue)	Data integration (ETL) and a catalogue of datasets/tables.
Golden dataset	Fixed test cases with expected properties used for every prompt change.
Graceful degradation	Reduced but useful service instead of errors.
GraphRAG (Graph-based RAG)	RAG that uses relationships between entities stored in a graph.
Grounding / groundedness	Whether an answer is supported by the retrieved sources.
Guardrail trace	Details of which policy intervened and why.
guardrailConfig / ApplyGuardrail	Attach a guardrail to a model call / check any text without a model call.
Hallucination	A confident answer that isn't supported by facts or sources.
Hierarchical chunking	Small 'child' chunks are searched; their larger 'parent' chunk is returned.
Hit rate (recall@k) / precision@k / MRR	Right chunk in top k / share of top k relevant / rewards rank 1.
HITL (Human-in-the-loop)	A person approves or reviews a step before it proceeds.
HNSW (Hierarchical Navigable Small World)	A fast algorithm for finding nearest vectors.
Hybrid search	Combines semantic and keyword (BM25) search.
Hyperparameters (epochs, learning rate, batch size)	Training settings for customisation jobs.

Term	Meaning
IAM (AWS Identity and Access Management)	Who (users, roles) can do what (actions) on which resources.
IAM Identity Center	Workforce single sign-on to AWS accounts and roles.
IAM role	An identity with permissions that services or apps assume, giving short-term credentials.
Idempotent	Repeating a request has the same effect as doing it once.
Identity-based vs resource-based policy	Identity policy is attached to the caller; resource policy is attached to the resource saying who may use it.
Implicit filtering	A model builds metadata filters from the user's question.
Implicit vs explicit deny	Implicit: nothing allowed it. Explicit: a Deny statement — always wins.
Inbound vs outbound auth	Inbound: who may call the agent/gateway. Outbound: how the agent calls other services.
Inference	Running a trained model to get an output (as opposed to training it).
inferenceConfig / toolConfig	Converse fields for inference parameters / tool definitions.
Input sanitisation	Cleaning untrusted input (hidden text, markup, length).
Input tagging	Marking which part of a prompt the guardrail should evaluate.
Instruction framework	Role/rules + guardrail policy + format spec controlling model behaviour together.
Intelligent Prompt Routing	Bedrock feature routing each prompt to the cheaper adequate model in a family.
Intent recognition	Classifying what a user wants (Comprehend custom classifier, Lex).
Invocation logging (Model invocation logging)	Optional Bedrock feature recording full prompts and responses to S3/CloudWatch.
InvokeModel	Model-specific Bedrock call; request format differs per provider.
Jailbreak	Persuading a model to ignore its rules.
JSON (JavaScript Object Notation)	Text format for structured data.
JSON Schema	Formal description of required JSON structure, used to force and validate outputs.
JSONL	File with one JSON object per line (batch inference input).
JSONPath / JSONata	Two expression languages Step Functions uses to read and transform data.
JWT (JSON Web Token)	A signed token proving who the caller is.
KB (Knowledge Base (Amazon Bedrock Knowledge Bases))	AWS's managed RAG service: ingests, chunks, embeds, stores and retrieves your documents.
Key policy	The resource policy on a KMS key saying who may use it.
KMS (AWS Key Management Service)	Creates and controls encryption keys; every use is auditable.
kNN (k-Nearest Neighbours)	Finding the k vectors closest to a query vector.
KPI	Key Performance Indicator.
KV cache	GPU working memory per conversation; grows with context and users.
Lake Formation (AWS Lake Formation)	Fine-grained table/column/row permissions for data lakes.
Lambda (AWS Lambda)	Runs your code on demand without servers (max 15 minutes per run).
Lambda authorizer	Custom token check protecting an API Gateway API.

Term	Meaning
Lambda response streaming	Lambda sends its response progressively (e.g. via function URLs).
Latency-optimised inference	Bedrock option for faster responses on supported models/Regions.
Lifecycle rule (S3 Lifecycle)	Automatically moves or deletes objects after N days.
LLM (Large Language Model)	A model trained on huge amounts of text that predicts the next words.
LLM-as-a-judge	Using a model to score another model's outputs against criteria.
LMI / TGI containers	Large Model Inference and Text Generation Inference serving containers.
Loose coupling	Systems interact through events, so each can change or fail independently.
LoRA / PEFT / adapter	Parameter-efficient fine-tuning: train small adapters on a frozen base model.
Macie (Amazon Macie)	Scans S3 to discover sensitive data such as PII.
Map state	Step Functions state running a step for every item, in parallel.
Map-reduce summarisation	Summarise chunks separately, then summarise the summaries.
Mapping template	API Gateway rule that transforms requests/responses.
Mask vs block (Guardrails PII actions)	Mask replaces the PII with a placeholder; block rejects the whole input/output.
Max tokens / stop reason max_tokens	Answer length limit; stop reason shown when the limit is reached.
MCP (Model Context Protocol)	Open standard for connecting AI apps to tools and data ('USB-C for AI tools').
Metadata filtering	Restricting retrieval to chunks whose attributes match (tenant, department).
microVM (Micro virtual machine)	A tiny, fast-starting isolated VM.
Modality / multimodal	Type of data a model handles; multimodal models accept e.g. text and images.
Model card / AI Service Card	Document of intended use, limitations, evaluation, risks.
Model distillation (teacher/student)	Large model's answers train a smaller model.
Model invocation logging	Bedrock logging of full prompt and response text to CloudWatch Logs / S3.
Model routing	Choosing the model per request to balance cost and quality.
Multimodal	Data beyond plain text: images, scans, audio, video.
NAT gateway (Network Address Translation gateway)	Lets private resources reach the internet outbound.
Neptune Analytics (Amazon Neptune Analytics)	Graph analytics database.
OAuth (Open Authorization)	Standard for letting an app act on a user's behalf without their password.
OCR (Optical Character Recognition)	Reading text from images or scanned pages.
OCU (OpenSearch Compute Unit)	Billing unit for OpenSearch Serverless capacity, charged hourly.
On-demand	Pay per token, shared capacity, no commitment.
OpenAPI (OpenAPI Specification)	A standard file describing a REST API's endpoints and parameters.
OpenAPI specification	Machine-readable API contract used for API-first development.
OpenSearch neural plugin	OpenSearch calls an embedding model itself at ingest and query time.
OpenSearch Serverless (Amazon OpenSearch Serverless)	Managed search/vector engine without clusters.
OpenTelemetry (OTel)	Open standard format for traces, metrics and logs.
Output diffing	Comparing outputs for the same input across runs or versions.

Term	Meaning
Output format specification	Exact required output shape: schema, fields, length.
p50 / p95 / p99	Latency percentiles (median, 95th, 99th).
p95 latency	Time that 95% of requests complete within.
Parallel / Map state	Step Functions states running branches or items concurrently.
Parameterised template / variable	Prompt with {{placeholders}} filled at runtime.
Pass / Choice / Task state (Step Functions states)	Pass: placeholder; Choice: branch; Task: do work.
PAT (Personal Access Token)	A long-lived token tied to one user (e.g. GitHub).
Permission boundary	IAM feature capping the maximum permissions a role can have.
PII (Personally Identifiable Information)	Data that identifies a person.
PoC (proof of concept)	Small, time-boxed build that validates feasibility, performance and value.
Pre-computation	Generating answers ahead of time for predictable requests.
Prefill	Starting the assistant's reply (e.g. '{') so the model continues in that format.
PrivateLink (AWS PrivateLink)	The technology behind interface VPC endpoints.
Programmatic model card	Model card created/updated by code in a pipeline.
Prompt attack	Attempt to override instructions or jailbreak.
Prompt caching / cache checkpoint	Reusing a processed static prompt prefix; the marker after the cached part.
Prompt chaining / decomposition	Fixed sequence of prompts, each output feeding the next.
Prompt compression	Same instructions in fewer tokens.
Prompt confusion	Model misbehaves because the rendered prompt is wrong or contradictory.
Prompt governance	Control of who changes prompts, approval, and audit.
Prompt Management / Prompt Flows	Bedrock prompt versioning and variants / visual prompt workflows.
Prompt variant	Alternative wording/model compared side by side in the console.
promptVariables	Converse parameter carrying variable values when modelId is a prompt ARN.
Provisioned Throughput / model unit (MU)	Reserved dedicated capacity billed hourly; MU = unit of capacity.
Quality gate	Pipeline step that fails a deployment below quality thresholds.
Quantisation	Storing weights/vectors with fewer bits to save memory.
Quarantine prefix	S3 location where invalid files are moved for review.
Query decomposition / rewriting	Split multi-part questions / make follow-ups standalone.
Query expansion / HyDE	Search with rephrasings / with a hypothetical answer.
Query rewriting / decomposition	Turning a follow-up into a standalone question / splitting a complex question.
RAG (Retrieval-Augmented Generation)	Search your own data first, then give the relevant passages to the model so it answers from them.
RBAC / ABAC	Access by role / access by attributes such as tags.
ReAct (Reason + Act)	The agent loop pattern: think → call a tool → observe → repeat.
Reasoning display	User-facing explanation of how an answer or decision was reached.
Reasoning path tracing	Inspecting each reasoning/tool step to find where logic fails.
Regex (Regular expression)	A text pattern for matching strings.

Term	Meaning
Regression testing	Proving a new version didn't break what used to work.
Reranker / rerank model	A second model that re-scores retrieved chunks so the best is first.
Responsible AI dimensions	Transparency, fairness, safety, privacy, veracity, controllability, governance.
REST / HTTP (Representational State Transfer)	Common style of web APIs over HTTP.
Retrieve vs RetrieveAndGenerate	Retrieve returns chunks only; RetrieveAndGenerate returns a full answer with citations.
RoC (Return of control)	Bedrock Agents hands the tool call back to your application to execute.
Route 53 failover	DNS routing that sends users to a healthy Region.
RRF (Reciprocal Rank Fusion)	A method of merging two ranked result lists into one.
S3 (Amazon Simple Storage Service)	Object storage for files of any kind.
S3 Object Lock (WORM)	Write once, read many: objects can't be changed or deleted for a period.
Safety classifier	Model or service that labels input/output as safe or harmful.
SageMaker AI / JumpStart	AWS ML platform; JumpStart = its pre-trained model catalogue; host on chosen instances.
SageMaker Clarify	AWS tool for bias detection and explainability.
SageMaker Data Wrangler	Visual data preparation with a data quality and insights report.
SageMaker Ground Truth / Amazon A2I	Labelling workforce service / human review loops.
SageMaker Model Monitor	Detects data, quality and bias drift for hosted models.
SageMaker Model Registry	Catalogue of model versions with approval status.
SageMaker Pipelines / CodePipeline	Automated ML / CI-CD pipelines.
SageMaker Processing job	Managed batch compute running your own preprocessing container.
SCP (Service Control Policy)	Organisation-wide guardrail capping what any member account (even admins) can do.
SDK (Software Development Kit)	Libraries for calling a service from code.
Secrets Manager (AWS Secrets Manager)	Stores and rotates secrets (passwords, API keys).
Semantic cache	Cache matched by embedding similarity of questions.
Semantic drift	Answers' meaning shifting versus a baseline for the same questions.
Semantic search	Search by meaning using vectors.
Semantic similarity verification	Comparing answer claims with source chunks by embedding similarity.
Sensitive information filter	PII detection with block or mask, plus custom regex.
Server-sent events (SSE)	One-way server-to-browser stream over HTTP.
Service Quotas	Console for viewing and raising account limits.
Shard / replica	Piece of an index on a node / copy of a shard for reads and availability.
Sidecar .metadata.json	File next to a KB document holding its metadata attributes.
SNS (Amazon Simple Notification Service)	Pub/sub notifications (email, SMS, HTTP).
Source attribution / citations	Showing which documents an answer came from.
Speaker diarization	Labelling who spoke when in a recording.

Term	Meaning
SQS (Amazon Simple Queue Service)	Message queue between components.
SQS / DLQ	Simple Queue Service; dead-letter queue for messages that keep failing.
SSE (Server-Side Encryption)	The service encrypts data when storing it.
SSE-S3 / SSE-KMS	SSE-S3: keys managed by S3. SSE-KMS: encryption using a KMS key you can control.
Stage variable	Per-stage setting in API Gateway (e.g. dev vs prod).
Standard vs Express (Step Functions workflow types)	Standard: up to 1 year, full history. Express: up to 5 minutes, high volume.
StartAsyncInvoke	Bedrock native async call for slow generation (e.g. Nova Reel video) → S3.
Step Functions (AWS Step Functions)	Visual workflows that chain steps with retries and history.
Stop sequence	String that ends generation when produced.
Stopping condition	Limit that ends an agent loop (iterations, budget, time).
Strands Agents	AWS's open-source Python SDK for building agents.
Streaming / TTFT	Sending tokens as generated; time to first token = perceived latency.
Structured input components	Delimited blocks (XML-style tags) separating rules, data and question.
Subnet (private)	A slice of a VPC with no direct internet route.
Supervisor / router	Supervisor: delegates to several agents and combines. Router: sends to exactly one.
Swarm	Peer agents hand work to each other without a central coordinator.
Sync / ingestion job	The process that reads a data source and updates the KB index.
Synchronous / asynchronous	Caller waits for result / result delivered later.
Systems Manager Parameter Store	AWS store for configuration values and secrets.
Task completion rate	Share of agent tasks fully and correctly finished.
Temperature	Setting controlling randomness of output; low = more deterministic.
Tensor parallelism	Splitting one model across several GPUs.
Text-to-SQL	Model writes SQL; database computes exact results.
Textract (Amazon Textract)	Extracts text, forms and tables from scans (OCR).
Throttling (ThrottlingException)	Error when you exceed a rate limit.
ThrottlingException (HTTP 429)	Over quota; retry with backoff.
Titan Multimodal Embeddings	Embedding model placing images and text in one vector space.
Token	A chunk of text (roughly ¾ of a word) that models read and write; billing and limits count tokens.
Token usage / token counting	Input/output token counts returned per call; estimating tokens before sending.
Token-level redaction	Masking individual sensitive tokens (PII) in text and logs.
Tool / function calling	The model requests a named tool with arguments; your code runs it.
toolChoice	Converse setting that forces the model to call a specific tool.
toolUse / toolResult	Converse API blocks: the model's tool request, and your reply with the output (matched by toolUseId).
Top-k	Sample only from the k most likely tokens.

Term	Meaning
Top-p (nucleus sampling)	Sample only from top tokens whose probabilities add up to p.
TPM / RPM (Tokens / Requests Per Minute)	Rate limits on model usage.
Transit Gateway	Hub connecting many VPCs and on-premises networks.
TSTALIASID (Test alias ID)	Built-in alias that points to the DRAFT version, for testing.
TTFT / tokens per second	Time to first token / generation speed.
TTL (Time To Live)	DynamoDB attribute that expires items automatically.
TTL / cache invalidation	Time-to-live for cached items / removing them when data changes.
Usage plan	API Gateway per-client throttling and quota tied to API keys.
Usage plan / API key	API Gateway per-client quotas and throttling.
ValidationException	Invalid request (e.g. input too long); don't retry, fix it.
Vector	The list of numbers produced by an embedding model.
VPC (Virtual Private Cloud)	Your private, isolated network inside AWS.
VPC endpoint / VPCE (VPC Endpoint)	A private connection from your VPC to an AWS service, no internet.
waitForTaskToken	Step Functions pattern that pauses until someone returns a task token.
Webhook	HTTP callback a third-party system sends when an event happens.
WebSocket API	API Gateway persistent two-way connections.
Word filters / managed profanity list	Exact-match blocked words; AWS-maintained profanity list.
WORM / Object Lock (Write Once, Read Many)	Data can't be changed or deleted for a set period.
X-Ray / Transaction Search (AWS X-Ray, CloudWatch Transaction Search)	Distributed tracing; Transaction Search must be enabled for AgentCore traces.
Zero-ETL integration	Managed sync between services without pipeline code (e.g. DynamoDB → OpenSearch).

Last word
You've covered all 98 skills. Trust the method: requirements list → strike failing options → managed, least-privilege, automated answer that satisfies every constraint. Good luck, 1:30 PM.